

MovementLogger — BLE FileSync button reference

Quick reference for the BLE FileSync panel in MovementLogger. Each button, what it does, when to use it, and what happens behind the scenes. May 2026.

Where to find it: Open MovementLogger, click **BLE FileSync (download from PumpTsueri)** to expand the panel. PIN is `123456` on first connect.

Connection lifecycle

Button	Enabled when	What it does
Scan	Disconnected (and not currently scanning)	Starts a 5-second BLE scan. Any peripheral advertising as <code>PumpTsueri</code> appears under <i>Discovered</i> . Click the device row that appears to bring up the Connect button. Re-clicking Scan clears the previous results.
Connect	After Scan found a device	Pairs and connects to the box. macOS / Linux / Windows pops a Bluetooth permission dialog and a PIN dialog the first time — enter <code>123456</code> . The OS remembers the pairing; subsequent connects skip the PIN. On success the file list auto-loads.
Disconnect	Connected	Cleanly tears down the BLE link (sends <code>LL_TERMINATE_IND</code> so the box can immediately accept the next host). The GUI returns to Idle state — Scan becomes available again. Box keeps advertising and is reachable within ~1 second.

Don't `kill -9` the GUI. A hard kill skips the graceful disconnect — the box's BLE link stays "occupied" until the supervision timeout (10–30 s on macOS), during which a fresh GUI launch can't connect. Close via **Cmd-Q** (red traffic light on macOS) for a clean exit, or simply Disconnect first.

File operations

Button	Enabled when	What it does

Refresh file list	Connected	Sends <code>0P_LIST (0x01)</code> to the box. Each file on the SD root comes back as a row with its size. Rows are split into two groups: • Sensor — <code>Sens*.csv</code> , <code>Gps*.csv</code> , <code>Bat*.csv</code> , <code>Mic*.wav</code> . These are session data and default-ticked. • Debug — <code>FW_INFO_vN.TXT</code> , <code>CHKWN.TXT</code> , the <code>Error_Log_Pump_Tsueri_*.log</code> error log, plus any macOS AppleDouble files (<code>.*_*</code>). Default-unticked. Each group header has a Tick all / Untick all toggle.
Download selected	Connected, at least one row ticked and not yet downloaded	Reads all ticked, not-yet-downloaded files one at a time (serial queue — the box can only stream one file at a time). The button shows <i>Download selected (N queued)</i> while running. Each row gets: • A live progress bar (1.23 / 5.00 MB) updating every ~4 KB received. • A green  downloaded badge on completion. Saved to the path in the Save to: field. <code>Sens*.csv</code> and <code>*_gps.csv</code> auto-fill the visualization form's top-level slots so you can hit Generate immediately. The same file can be re-downloaded by clicking <i>Refresh file list</i> first.
 (per row)	Connected, that row is not currently downloading	Sends <code>0P_DELETE (0x03)</code> for that single file. On success (<code>0x00</code> OK reply) the row vanishes from the list. Other replies surface in the Log panel: BUSY (logging in progress, send <code>STOP_LOG</code> first), NOT_FOUND, IO_ERROR. Be deliberate — there is no confirmation dialog.

Logging sessions

Button	Enabled when	What it does
STOP_LOG	Connected	Sends <code>0P_STOP_LOG (0x04)</code>. If a logging session is in progress on the box, this gracefully closes <code>SensNNN.csv</code> , <code>GpsNNN.csv</code> , etc. (so they're complete and downloadable). No reply from the box — the GUI logs <i>STOP_LOG sent</i> immediately. After a few seconds you can <i>Refresh file list</i> to see the closed files.
1800 s	Connected	Session duration in seconds (1–86400). Click and drag to scrub, or click and type. Drives the next Start session. Default is 1800 s (30 minutes). For short tests drop to 60 s.
Start session	Connected	Sends <code>0P_START_LOG (0x05)</code> plus the duration. The box writes a magic value to its backup register and immediately reboots into LOG mode (the BLE stack does not run in LOG

		mode — it's mutually exclusive with SD logging on this hardware). The BLE connection drops; the GUI flips to Idle. The box logs for the requested duration, then auto-reboots back into BLE mode and advertises <code>PumpTsueri</code> again.
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While a session is running the GUI shows an amber banner above the action buttons: *"Logging in progress — M min SS s left ... Box is silent, won't appear in Scan."* The countdown ticks every ~500 ms and clears itself automatically once the deadline expires (plus ~8 s for BLE bring-up). Once cleared, click Scan and the box reappears.

To abort a session early short-press the physical user button on the box. `fx_thread` sees the button event and closes the files cleanly; in some firmware versions the box also reboots back into BLE mode.

Sidebar elements

Element	What it does
Save to:	Where downloads land on your computer. Defaults to <code>./csv/</code> relative to the GUI binary. Edit inline or click Browse... to pick a folder. The same path is used for every download in the session.
Log panel (bottom)	Scrolling history of every BLE state transition: <i>LIST sent, reading X (N B)...</i> , <i>saved /path/to/X</i> , <i>DELETE X sent</i> , errors, etc. Fully selectable: click-drag to highlight, Cmd-A select all, Cmd-C copy. The Copy all button in the panel header dumps the whole buffer to the clipboard in one click.
Window size	The Log panel grows vertically with the window — drag the bottom-right corner down to see more lines without scrolling.

Status bytes you might see in the Log

Code	Meaning	What to do
<code>0x00</code>	OK	Operation succeeded. (For READ/DELETE you'll see the  badge / row vanish.)
<code>0xB0</code>	BUSY	Box is currently logging — close files first with STOP_LOG , then retry.
<code>0xE1</code>	NOT_FOUND	File is in the LIST but the box can't open it. Usually FAT corruption on the SD card — pull the card, copy off what you need, reformat as FAT32.
<code>0xE2</code>	IO_ERROR	Same diagnosis as NOT_FOUND but on the write side (DELETE). Reformat the SD card.

0xE3	BAD_REQUEST	Malformed opcode or filename — bug in the GUI. Please report.
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Typical workflow

1. **Scan** → wait 5 s → click discovered PumpTsueri → **Connect** (PIN 123456).
2. File list auto-loads. Adjust ticks if needed.
3. **Download selected**. Wait for  badges.
4. To record more data: set **1800 s** (or your preferred duration) → **Start session**. Watch the amber countdown banner. When it clears, the box is back.
5. **Scan** → **Connect** → **Refresh file list** → **Download selected** on the new SensMMN.csv row.
6. **Disconnect** when done, or just **Cmd-Q** the app (it'll Disconnect automatically).

Where the code lives

- **Worker / BLE state machine:** Utilities/rust/stbox-viz-gui/src/ble.rs
- **UI / button handlers:** Utilities/rust/stbox-viz-gui/src/main.rs (ui_ble_panel , pump_ble_events)
- **Firmware GATT service:** Projects/STEVAL-MKBOXPRO/Applications/Rev_C/SDDataLogFileX/Core/Src/ble_filesync.c
- **Wire protocol reference:** [BLE FileSync.pdf](#)